

Data Analysis Project: Diabetes Prediction with Bayesian

Sandipan Dey

Executive summary

In this project, we shall try to understand how diabetes is dependent on a given a set of diagnostic measurements. More specifically, our objective will be:

1. Use the diagnostic measurements as predictor variables and *estimate* the coefficients for the predictors under Bayesian settings
2. *Predict* the response variable, i.e., whether a patient is *diabetic* or not, given those measurements and computed coefficients
3. Construct *Bayesian Credible Intervals (CI)* for the coefficients to quantify uncertainty and make probabilistic claims on the coefficients.
4. Compute the *Posterior Predictive Distribution* and use it to compute *CI* for the predictions and make probabilistic claims on the predictions for different patients.
5. We shall use base `R` and `rjags` library functions for the implementation.

Introduction

Predicting whether a patient is diabetic from her medical diagnosis will be the goal of this project. Under Bayesian settings, we shall impose *priors* on the coefficients corresponding to the predictors (diagnostic measurements) and use *MCMC* sampling to compute the *posterior* distributions, verifying that the Markov chains converge to the target stationary distribution with *diagnostic tests*. We shall try to fit a few different models and use *model checking / selection* to find the best model (with *DIC*), along with that we shall use *posterior predictive distribution* to answer to a few questions (test *hypotheses*) related to prediction of the outcomes, we shall also construct *Bayesian credible intervals* to quantify the uncertainty for the coefficient estimates and also for our prediction. As we shall see, few of the predictor variables we shall find as statistically significant.

Data

We shall use the Diabetes dataset from Kaggle: <https://www.kaggle.com/datasets/mathchi/diabetes-data-set> (<https://www.kaggle.com/datasets/mathchi/diabetes-data-set>). This dataset is originally from the National Institute of Diabetes and Digestive and Kidney Diseases. Several constraints were placed on the selection of these instances from a larger database. In particular, all patients here are females at least 21 years old of Pima Indian heritage. Each row corresponds to the measurements corresponding to a patient (there are 768 rows in the dataset), containing 8 **predictor** variables listed and described below (which can be used to predict the response variable), with a single **response** variable named *Outcome* (which is 1 if the patient is diabetic, 0 otherwise).

- Pregnancies: Number of times pregnant
- Glucose: Plasma glucose concentration a 2 hours in an oral glucose tolerance test
- BloodPressure: Diastolic blood pressure (mm Hg)
- SkinThickness: Triceps skin fold thickness (mm)
- Insulin: 2-Hour serum insulin (mu U/ml)
- BMI: Body mass index (weight in kg/(height in m)^2)
- DiabetesPedigreeFunction: Diabetes pedigree function
- Age: Age (years)
- Outcome: Class variable (0 or 1)

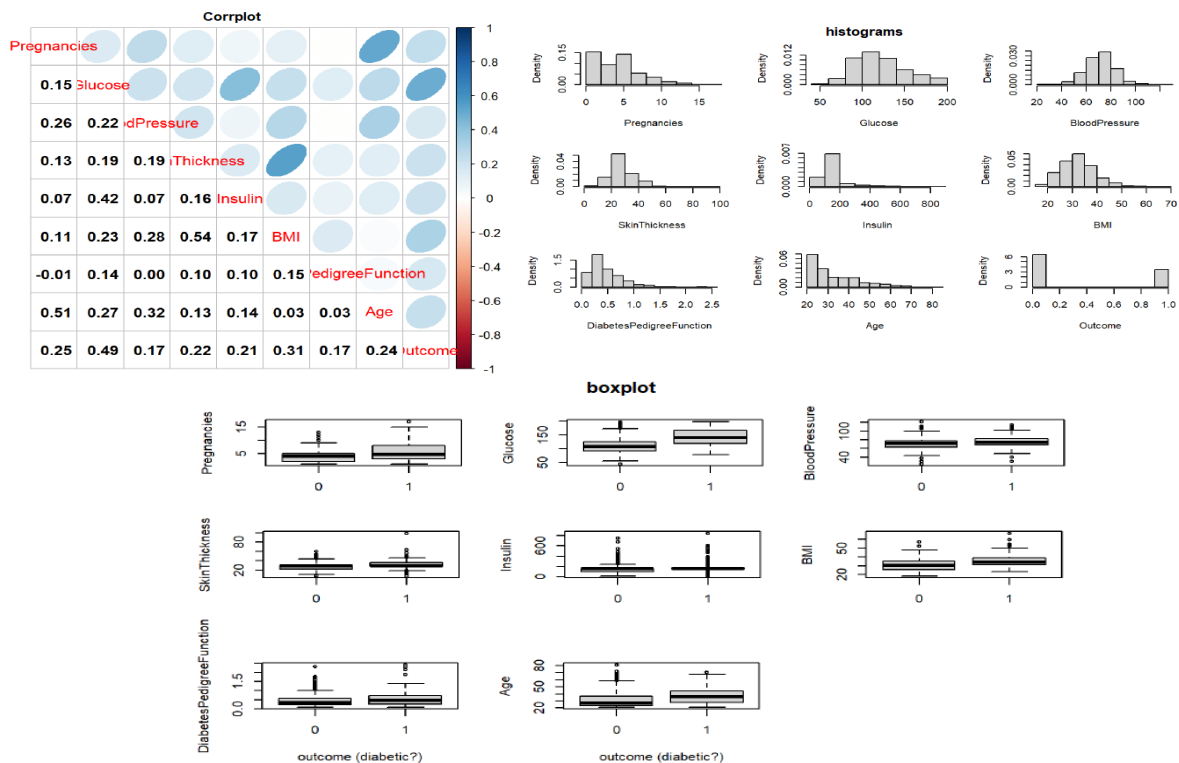
	Pregnancies <int>	Glucose <int>	BloodPressure <int>	SkinThickness <int>	Insulin <int>	BMI <dbl>	DiabetesPedigreeFunction <dbl>	Age <dbl>	Outcome <int>
1	6	148	72	35	0	33.6	0.627	50	1
2	1	85	66	29	0	26.6	0.351	31	0
3	8	183	64	0	0	23.3	0.672	32	1
4	1	89	66	23	94	28.1	0.167	21	0
5	0	137	40	35	168	43.1	2.288	33	1
6	5	116	74	0	0	25.6	0.201	30	0

Exploratory Analysis & Data Cleaning

- First we omit the **NA** values, if any, from the dataset.
- We observe that many of the predictors (e.g., *BloodPressure*, *SkinThickness*, *Insulin*) contain quite a few *zero* values (which appears to be impossible for the measurements, hence), we suspect they are noise.
- We first compute what percentage of values for a predictor contain inappropriate 0 values and find that there are quite significant (as shown below), so we can't drop the corresponding observations, since they will reduce the data size a lot.
- Instead, we use **mean imputation** technique to replace the *zero* values for a predictor with the *mean* of its non-zero values.
- Now, let's compute the **summary** of the dataset and visualize the correlation between the variables (separately for *diabetes* and *non-diabetes* observations, denoted by the colors red and blue, respectively), as shown below.

##	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
##	Min. : 1.000	Min. : 44.00	Min. : 24.00	Min. : 7.00	Min. : 14.0	Min. : 18.20	Min. : 0.0780	Min. : 21.00	Min. : 0.000
##	1st Qu.: 2.000	1st Qu.: 99.75	1st Qu.: 64.00	1st Qu.: 25.00	1st Qu.: 121.5	1st Qu.: 27.50	1st Qu.: 0.2437	1st Qu.: 24.00	1st Qu.: 0.000
##	Median : 4.495	Median : 117.00	Median : 72.20	Median : 29.15	Median : 155.5	Median : 32.40	Median : 0.3725	Median : 29.00	Median : 0.000
##	Mean : 4.495	Mean : 121.69	Mean : 72.41	Mean : 29.15	Mean : 155.5	Mean : 32.46	Mean : 0.4719	Mean : 33.24	Mean : 0.349
##	3rd Qu.: 6.000	3rd Qu.: 140.25	3rd Qu.: 80.00	3rd Qu.: 32.00	3rd Qu.: 155.5	3rd Qu.: 36.60	3rd Qu.: 0.6262	3rd Qu.: 41.00	3rd Qu.: 1.000
##	Max. : 17.000	Max. : 199.00	Max. : 122.00	Max. : 99.00	Max. : 846.0	Max. : 67.10	Max. : 2.4200	Max. : 81.00	Max. : 1.000

- let's plot the correlation between the predictors and the outcome, with darker color representing higher correlation.
- Next let's find out whether the distributions of the predictors given *Outcome=0* and *Outcome=1* differ significantly using the following *boxplot*. As can be seen from below, the distribution of the predictor variable *SkinThickness* is almost same when conditioned on different values of the *outcome*, so it's likely not to be a good predictor.
- Also, let's visualize the predictor distributions from the data using histograms (we can see some of the predictors in the data, e.g. *Age* are very much *right-skewed*).



Model

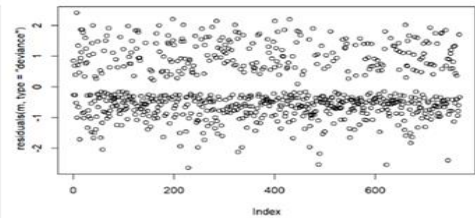
- We shall first fit the *glm* (generalized linear model) in R with `family=binomial`, with *Outcome* as the response and the rest of the variables as predictors. Plot the residuals (diagnostic plot) to see if there is any pattern and whether the model assumptions are met. Note the values of the coefficients computed, the statistically significant ones. Finally, predict using the model fitted and compute the prediction accuracy along with the *confusion matrix*.
- Next, we shall fit models with *rjags* with *Bernoulli* likelihood with *logit* link function. First, we shall fit the following model and Gaussian prior (with zero mean, sd 5) on the intercept, along with *Laplace* (double exponential) prior with zero mean and unit variance) on the remaining coefficients, to find posterior with sparse coefficients.
- We shall use 3 independent Markov chains for mixing, reject the first 1000 samples from the *burn-in* period and simulate for another 5000 iterations. use *trace plot* to check the convergence (autocorrelation) as diagnostic.
- Also, in order to examine convergence of the *Markov chains* to *stationary* (target posterior) distribution, we shall use the *Gelman-Brooks-Rubin* diagnostic to verify if the Markov chains don't differ much / end in the same state-space after convergence (also check the effective sample size).
- We shall use the *Deviance information criterion* (*DIC*) for model selection (the lower the value, the better the model fit) and to compute the effective parameter size of the problem.
- We shall see that the posterior distribution of a few coefficients includes zero, so they are likely to be less important as predictors. we shall get rid of them and refit the model. this time with Gaussian prior on all the coefficients.
 - likelihood: $y_i | x_i, \beta \stackrel{ind}{\sim} \text{Bern} \left(\frac{1}{1 + e^{-(\beta_0 + \sum_{j>1} \beta_j x_{ij})}} \right)$
 - priors: $\beta_j \sim N(0, 5^2)$
- Finally, we shall fit a third model using all the eight predictors and placing *flat normal* priors on them. We shall compare the models using *DIC* and select the best model.

Results

Logistic Regression with GLM

- The predictors `Pregnancies`, `Glucose`, `BMI`, `DiabetesPedigreeFunction` and `BloodPressure` are statistically significant (with $p\text{-value} < 0.05$, the null hypothesis can be rejected at 95% level of significance)
- The residual plot looks okay and does not show any pattern.
- Interpretation of the coefficients: e.g., increase in `Glucose` by 1 unit increases the *log odds* for diabetes by 0.037, whereas decrease in `BloodPressure` by 1 unit increases the *log odds* for diabetes by 0.01046
- The prediction has accuracy of 77.73%.

```
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -9.0705975  0.8148288 -11.132 < 2e-16 ***
## Pregnancies    0.1406610  0.0359271   3.915 9.03e-05 ***
## Glucose        0.0372466  0.0038779   9.605 < 2e-16 ***
## BloodPressure  -0.0104619  0.0086383  -1.211 0.22585
## SkinThickness  0.0038213  0.0131661   0.290 0.77164
## Insulin       -0.0007215  0.0011808  -0.611 0.54115
## BMI           0.0890451  0.0177613   5.013 5.35e-07 ***
## DiabetesPedigreeFunction 0.8541331  0.2981939   2.864 0.00418 **
## Age           0.0152995  0.0091727   1.668 0.09533 .
```



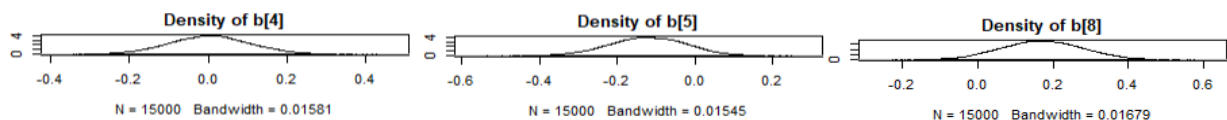
Bayesian Logistic Regression with RJags

Model 1

For the model with *double exponential (Laplace)* prior on the coefficients,

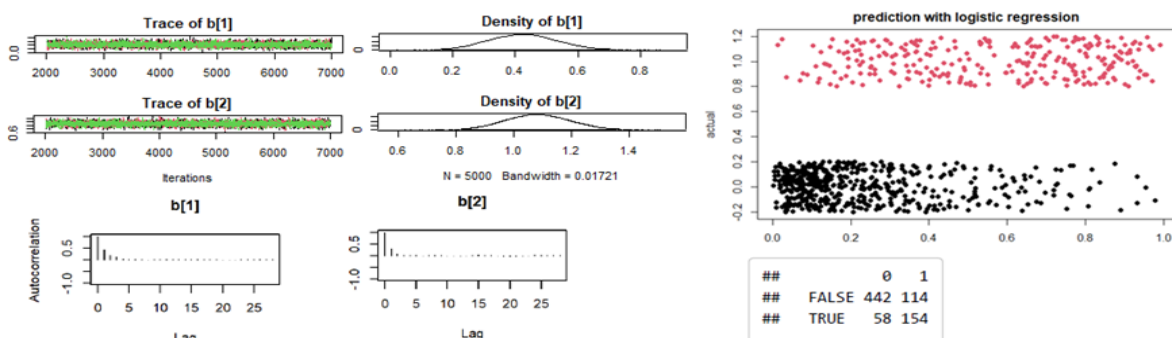
- The trace plots for the MCMC looks okay, indicating convergence.
- The Gelman diagnostic looks okay too, the values are close to 1, as expected.
- The coefficients for `SkinThickness`, `Insulin` and `Age` includes zero in posterior distribution, we can ignore them.
- The DIC for the model is 740.7

```
mod1_string = " model {
  for (i in 1:length(outcome)) {
    outcome[i] ~ dbern(p[i])
    logit(p[i]) = int + b[1]*Pregnancies[i] + b[2]*Glucose[i] + b[3]*BloodPressure[i] + b[4]*SkinThickness[i]
                  + b[5]*Insulin[i] + b[6]*BMI[i] + b[7]*DiabetesPedigreeFunction[i] + b[8]*Age[i]
  }
  int ~ dnorm(0.0, 1.0/25.0)
  for (j in 1:8) {
    b[j] ~ ddexp(0.0, sqrt(2.0)) # has variance 1.0
  }
}
```



Model 2

- Ignoring `SkinThickness`, `Insulin` and refitting the model with flat normal prior (with zero mean, sd 5) on each of the predictor coefficients results in *DIC* of 739.5, a slight improvement from the previous model.
- The trace plots look okay, the Markov chains seem to have converged, with the autocorrelation plot supporting it. Effective sample sizes for the coefficients are quite large too.
- The model is used for prediction and it gives 77.60% accuracy.



Model 3

- Using all eight of the predictors and refitting the model with flat normal prior (with zero mean, sd 5) on each of the predictor coefficients results in *DIC* of 741.3, a slight deterioration from the previous model.
- The trace plots look okay, the Markov chains seem to have converged, with the autocorrelation plot supporting it. Effective sample sizes for the coefficients are quite large too.
- The model is used for prediction and it gives 78.25% accuracy (on the same dataset it's trained on).

Model 2 turns out to be the **best model** among the above three, in terms of **DIC** (lowest).

Bayesian Credible Intervals (CI) for the Coefficients

The *Highest Posterior Density intervals* for the coefficients are shown below:

- e.g., the estimated coefficient for the predictor `Glucose` lies in the interval `[0.9245, 1.3930]` with 95% probability.

```
##           lower      upper
## b[1]  0.21011801  0.63532505
## b[2]  0.92446245  1.39304542
```

Make Probabilistic Statements

For example, let's try to answer the question: what's the probability that the effect size for `DiabetesPedigreeFunction` is greater than that of the predictor `Age`?

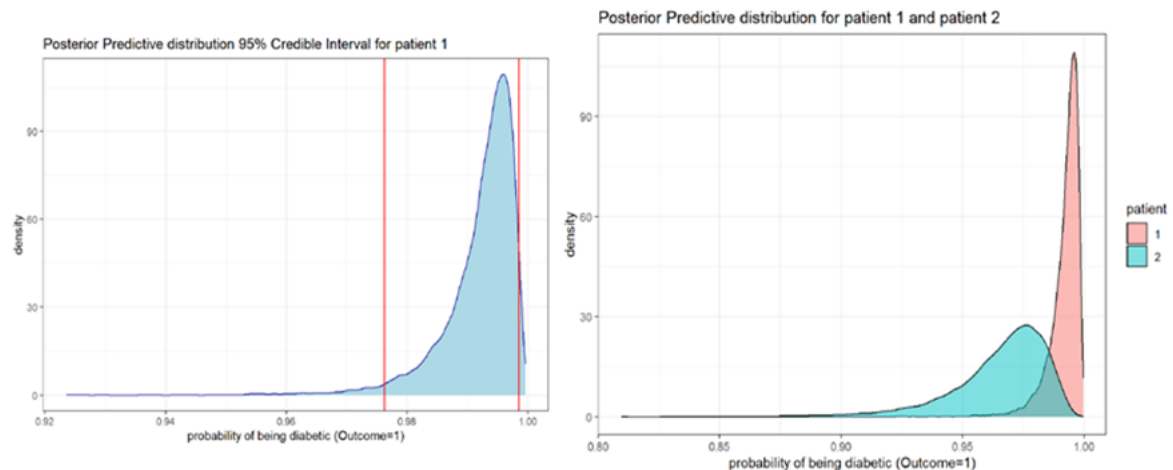
We can answer by conditioning on the samples obtained from simulation and then taking mean: the probability is 0.76

Posterior Predictive Distribution

- let's consider the following two new patients with the following diagnostic measurements for the predictors.

	Pregnancies <dbl>	Glucose <dbl>	BloodPressure <dbl>	SkinThickness <dbl>	Insulin <dbl>	BMI <dbl>	DiabetesPedigreeFunction <dbl>	Age <dbl>
patient 1	0	300	80	30	100	33	0.30	40
patient 2	0	250	82	29	90	35	0.25	50

- Now, let's use the MCMC samples drawn to construct the posterior predictive distribution for patient one and construct 95% credible interval with *quantile*. Hence, with 95% of the time the probability that the first patient is diabetic will be in the interval `[0.9708372, 0.9978117]`.
- Also, let's compute the probability that the second patient has higher probability of having diabetes than the first patient. As can be seen, it can also be computed from the MCMC samples and the result is 0.0084



Conclusions

Using Bayesian Logistic Regression, we could not only compute the point estimates for the predictors and use them to predict the outcome, but the main strength of the model had been to quantify uncertainty in terms of how confident we are in terms of the estimates and predictions (using posterior predictive distribution). We can see that all predictor variables other than `Insulin` and `BloodPressure`, increase in their values increase the log odds of diabetes. The model even allows us to make probabilistic statements in terms of comparing two patients' probability of being diabetic, given the values of the diagnostic measurements of the predictor variables.